USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 28 LAB - FMCW Range
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.2: I can calculate range resolution, unambiguous range, and Doppler shift for a given radar waveform.

Learning Objective 4.4: I can describe FMCW radar operation and configure an FMCW waveform on the ADALM-PHASER.

Learning Objective 4.5: I can process FMCW radar data to produce range, range-waterfall, and range-Doppler results.

This packet is the L28 Lab, in three parts, and you turn it in once, complete, before Lesson 29, with your notebook uploaded beside it. **Part 1, Predict**, is yours to do before class: the shaded Predicted cells of each table. **Part 2, Measure**, is your own kit's readings and your notebook's, in the white Measured cells of the same tables, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before class)

1. **Predict.** Fill in every shaded **Predicted** cell of the Setup table and Tables A, B, C, and E. Lesson 28's Pre-Lab frame names the frame that works each method, on a chirp no preset uses. Attach your work for one row of each table on your own paper. Use $c = 3.00 \times 10^8$ m/s.

    - **The waveforms.** The Phaser app's **Range (Basic)** preset sweeps $B = 500$ MHz in $T_s = 1$ ms from 12.145 GHz and samples the beat at $f_s = 0.6$ MSPS. **Range (Fast)** sweeps 500 MHz in 0.5 ms from 10 GHz at 5 MSPS. The recorded 250 MHz capture is Range (Basic) with half the bandwidth.
    - **The app.** It keeps the last 90% of each chirp, after the ramp settles. Its IF tone is 100 kHz: take that as f_0, the 0 m frequency, for every prediction.
    - **The readout.** The notebook reads widths with `radar_lab`'s Hann window, and every height relative to the noise floor of the same capture.
    - **The targets.** A plate at 1.00, 2.00, and 3.00 m by the tape, and the course's two corners 0.20 m apart near 2 m.

Part 2: Measure (your kit and the notebook, in class)

Follow the steps on the L28 page and fill in every white **Measured** cell, from your own kit and your own captures; the notebook's peak readout gives the frequencies. With no kit, use the app's simulator and the recorded captures, as the page's *No Hardware?* frame describes, and mark which cells are recorded. A dash marks a cell with nothing to fill.

Kit: _____ Recorded data? _____

Setup	**Predicted**	**Measured**
Samples per chirp, Range (Basic)		
Samples per chirp, Range (Fast)		

2. **Table A: the 0 m frequency** (Steps 3 to 5, and 17). Range (Basic), before the calibration. The app reads to the nearest FFT bin; the notebook reads the same capture to 0.01 kHz.

	Predicted	**App**	**Notebook**
Leakage, clear range (kHz)			
f_0, plate on the array (kHz)			

3. **Table B: static ranges** (Steps 7 to 9, 17, and 20). Range (Basic), calibrated. Fit a line through the three peak frequencies against the tape ranges.

		Peak frequency (kHz)	
Tape range	**Range in the app**	**Predicted**	**Measured**
1.00 m			
2.00 m			
3.00 m			
Beat per meter, the slope (kHz/m)	–		
Intercept (kHz)	–		
R from your f_0, 2 m capture (m)	–	–	

4. **Table C: chirp length and bandwidth** (Steps 10 to 12, and 17). Plate at 2.00 m. The kHz per meter is the beat above your notebook f_0 from Table A, over 2.00 m. The height is the change in the peak's height above the floor, relative to Range (Basic).

	Peak (kHz)	kHz per meter	Width 3 dB (m)	Height (dB)
Predicted Range (Basic)				–
Measured				–
Predicted Range (Fast)				
Measured				
Predicted 250 MHz				
Measured				

5. **Table D: the waterfall** (Steps 13 to 15). Your walk, from the app's Range Waterfall, and the course's recorded walk, played with Load Recording. Time for one leg is the frames over the frame rate; speed is the leg's length over its time.

	Your walk	`L28_walker.npz`
Nearest range of the trace (m)		
Farthest range of the trace (m)		
Frames for one leg		
Frame Rate (fps)		
Time for one leg (s)		
Walking speed (m/s)		

6. **Table E: windows and zero-padding** (Steps 18 and 19). Task (a) on your 2 m capture; task (b) on the course's two corners, 0.20 m apart.

Task (a)	Hann	Rectangular	
Strongest sidelobe (dBc)			–
Floor near 10 m (dBFS)			–

Task (b)	Predicted	No padding	4096 points
Peaks between 1.5 and 2.7 m			
Width 3 dB down (m)	–		

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

7. **Beat per meter.** Compare your Table B slope and intercept with your predictions and with your Table A frequencies. State what agrees and what does not, give the cause you think most likely for each difference, and say what measurement on this kit would test it.

8. **Three waveforms.** Compare the three waveforms of Table C with your predictions: the peak, the kHz per meter, the width, and the height. Which predictions held within what your measurement can resolve, which did not, and what explains each difference?

9. **The walk.** From your waterfall and the recorded one in Table D, report your walking speed with its uncertainty, and say what the waterfall doesn't explain about how you moved.

Documentation: _____